Blockchain Experiment:11

Aim - To implement a Hyperledger Fabric network, deploy chaincode, and perform asset creation, querying, and transfer operations.

Theory -

Hyperledger Fabric Network Setup

- Consortium & Membership Service Provider (MSP): Fabric is permissioned, meaning only authorized organizations can join. MSPs define identities and roles.
- Peers: Nodes that maintain the ledger and endorse transactions. Each organization typically runs peers.
- Orderer: The ordering service packages endorsed transactions into blocks and delivers them to peers.
- Ledger: Composed of a blockchain (immutable transaction log) and a state database (current values).
- Channels: Private sub-networks where a subset of organizations transact confidentially.

Chaincode (Smart Contracts)

- Chaincode defines the business logic for asset management.
- Written in Go, Java, or Node.js.
- Deployed to peers, then instantiated on a channel.
- Functions typically include createAsset, queryAsset, transferAsset, and deleteAsset.

Transaction Flow

1. Proposal Creation: A client application submits a transaction proposal (e.g., create asset).
2. Endorsement: Peers simulate the transaction using chaincode and return signed results.
3. Ordering: The orderer sequences endorsed transactions into blocks.
4. Validation & Commit: Peers validate endorsements, check for conflicts, and commit the block to the ledger.

Asset Operations

- Asset Creation:
  - Chaincode function createAsset stores asset attributes (ID, owner, value) in the ledger's state database.
- Asset Querying:
  - queryAsset retrieves asset details by key. Rich queries are possible if CouchDB is used as the state database.
- Asset Transfer:
  - transferAsset updates the owner field of an asset. The transaction is endorsed, ordered, and committed, ensuring immutability and consensus.

Key Principles

- Immutability: Once committed, transactions cannot be altered.
- Consensus: Endorsement policies ensure multiple organizations agree before a transaction is valid.

- Privacy: Channels and private data collections allow selective visibility.
- Trust: Identities are cryptographically verified via MSPs.

In essence, Hyperledger Fabric provides a modular, permissioned blockchain framework where chaincode governs asset lifecycles. The theory emphasizes identity-driven trust, modular architecture, and consensus-based validation — ensuring secure creation, querying, and transfer of assets across organizations.

Prerequitise: Check if WSL is Installed

Open **PowerShell (Run as Administrator)** and run:
wsl --list --verbose
If it shows **no distributions**, you must install Ubuntu.

Install Ubuntu for WSL
Run this command in **PowerShell (Admin)**:
wsl --install -d Ubuntu

This installs:
- Ubuntu
- WSL2 kernel

The installation may take a few minutes.
Restart Your Computer

After installation finishes, **restart your PC**.
Open Ubuntu
Open:
Start Menu → Ubuntu

It will ask you to create:
Username

Password

Now Ubuntu is ready.

User myuser
Pass : user123

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\WINDOWS\system32> wsl --install
>>
Ubuntu is already installed.
Launching Ubuntu...
Welcome to Ubuntu 24.04.1 LTS (GNU/Linux 5.15.153.1-microsoft-standard-WSL2 x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Fri Apr  3 12:42:09 UTC 2026

  System load:  0.83                Processes:             71
  Usage of /:   0.1% of 1006.85GB   Users logged in:       0
  Memory usage: 5%                  IPv4 address for eth0: 172.25.13.113
  Swap usage:   0%

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
   just raised the bar for easy, resilient and secure K8s cluster deployment.

   https://ubuntu.com/engage/secure-kubernetes-at-the-edge
```

```
 Swap usage:   0%

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
   just raised the bar for easy, resilient and secure K8s cluster deployment.

   https://ubuntu.com/engage/secure-kubernetes-at-the-edge

This message is shown once a day. To disable it please create the
/root/.hushlogin file.
root@snehal:~# adduser myuser
info: Adding user `myuser' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `myuser' (1000) ...
info: Adding new user `myuser' (1000) with group `myuser (1000)' ...
info: Creating home directory `/home/myuser' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for myuser
Enter the new value, or press ENTER for the default
        Full Name []: snehal
        Room Number []:
        Work Phone []:
        Home Phone []:
        Other []:
Is the information correct? [Y/n] y
info: Adding new user `myuser' to supplemental / extra groups `users' ...
info: Adding user `myuser' to group `users' ...
root@snehal:~#
```

```
Is the information correct? [Y/n] y
info: Adding new user `myuser' to supplemental / extra groups `users' ...
info: Adding user `myuser' to group `users' ...
root@snehal:~# usermod -aG sudo myuser
root@snehal:~# su - myuser
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

Welcome to Ubuntu 24.04.1 LTS (GNU/Linux 5.15.153.1-microsoft-standard-WSL2 x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Fri Apr  3 12:46:19 UTC 2026

  System load:  0.02                Processes:             31
  Usage of /:   0.1% of 1006.85GB   Users logged in:       1
  Memory usage: 5%                  IPv4 address for eth0: 172.25.13.113
  Swap usage:   0%

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
   just raised the bar for easy, resilient and secure K8s cluster deployment.

   https://ubuntu.com/engage/secure-kubernetes-at-the-edge

This message is shown once a day. To disable it please create the
/home/myuser/.hushlogin file.
myuser@snehal:~$
```

PHASE 1 — SYSTEM SETUP

Step 1: Update system

code:
sudo apt update
sudo apt install curl build-essential git -y

Step 2: Install Go 1.18.10

code:
wget https://go.dev/dl/go1.18.10.linux-amd64.tar.gz
sudo rm -rf /usr/local/go && sudo tar -C /usr/local -xzf go1.18.10.linux-amd64.tar.gz
echo 'export PATH=$PATH:/usr/local/go/bin' >> ~/.code:rc
echo 'export GOPATH=$HOME/go' >> ~/.code:rc
echo 'export PATH=$PATH:$GOPATH/bin' >> ~/.code:rc
source ~/.code:rc
go version

```
myuser@snehal:~$ sudo apt update
[sudo] password for myuser:
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:4 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1576 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble/universe amd64 Packages [15.0 MB]
Get:7 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [252 kB]
Get:8 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21.5 kB]
Get:9 http://security.ubuntu.com/ubuntu noble-security/main amd64 c-n-f Metadata [10.7 kB]
Get:10 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [1167 kB]
Get:11 http://security.ubuntu.com/ubuntu noble-security/universe Translation-en [225 kB]
Get:12 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [74.2 kB]
Get:13 http://security.ubuntu.com/ubuntu noble-security/universe amd64 c-n-f Metadata [22.8 kB
```

```
myuser@snehal:~$ sudo apt install curl -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libcurl3t64-gnutls libcurl4t64
The following packages will be upgraded:
  curl libcurl3t64-gnutls libcurl4t64
3 upgraded, 0 newly installed, 0 to remove and 220 not upgraded.
Need to get 902 kB of archives.
After this operation, 0 B of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 curl amd64 8.5.0-2ubuntu10.8 [22
Get:2 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libcurl4t64 amd64 8.5.0-2ubuntu1
 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libcurl3t64-gnutls amd64 8.5.0-2
.8 [334 kB]
Fetched 902 kB in 2s (402 kB/s)
(Reading database ... 40788 files and directories currently installed.)
```

```
myuser@snehal:~$ sudo apt install build-essential -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  bzip2 cpp cpp-13 cpp-13-x86-64-linux-gnu cpp-x86-64-linux-gnu dpkg dpkg-dev fakeroot g++ g++-13
  g++-13-x86-64-linux-gnu g++-x86-64-linux-gnu gcc gcc-13 gcc-13-base gcc-13-x86-64-linux-gnu
  gcc-14-base gcc-x86-64-linux-gnu libalgorithm-diff-perl libalgorithm-diff-xs-perl
  libalgorithm-merge-perl libaom3 libasan8 libatomic1 libc-bin libc-dev-bin libc-devtools libc6
  libc6-dev libcc1-0 libcrypt-dev libde265-0 libdpkg-perl libfakeroot libfile-fcntllock-perl
  libgcc-13-dev libgcc-s1 libgd3 libgomp1 libheif-plugin-aomdec libheif-plugin-aomenc
  libheif-plugin-libde265 libheif1 libhwasan0 libisl23 libitm1 liblsan0 libmpc3 libquadmath0
  libstdc++-13-dev libstdc++6 libtsan2 libubsan1 libxpm4 linux-libc-dev locales lto-disabled-list make
  manpages-dev rpcsvc-proto
Suggested packages:
```

```
myuser@snehal:~$ curl -fsSL https://deb.nodesource.com/setup_18.x | sudo -E bash -
sudo apt install -y nodejs
```

```
myuser@snehal:~$ sudo snap install --classic --channel=1.18/stable go
2026-04-03T13:00:22Z INFO Waiting for automatic snapd restart...
go (1.18/stable) 1.18.10 from Canonical✓ installed
```

Install weftility

code:
npm install -g @hyperledgendary/weftility

Verify:

code:

weft --version

Expected: a version number, no errors.

```
myuser@snehal:~$ sudo npm install -g @hyperledgendary/weftility
npm warn deprecated inflight@1.0.6: This module is not supported, and leaks memory. Do not use it. Ch
 out lru-cache if you want a good and tested way to coalesce async requests by a key value, which is
h more comprehensive and powerful.
npm warn deprecated rimraf@3.0.2: Rimraf versions prior to v4 are no longer supported
npm warn deprecated glob@7.2.3: Old versions of glob are not supported, and contain widely publicized
curity vulnerabilities, which have been fixed in the current version. Please update. Support for old
sions may be purchased (at exorbitant rates) by contacting i@izs.me

added 158 packages in 34s

22 packages are looking for funding
  run `npm fund` for details
```

Step 3: Install Docker Compose v1.29.2

code:
sudo curl -L "https://github.com/docker/compose/releases/download/1.29.2/docker-compose-$(uname -s)-$(uname -m)" -o /usr/local/bin/docker-compose
sudo chmod +x /usr/local/bin/docker-compose
docker-compose version

Expected: docker-compose version 1.29.2

Before continuing, open Docker Desktop on Windows → Settings → Resources → WSL Integration → Enable for your Ubuntu distro → Apply & Restart. Then verify:

code:
docker version

```
myuser@snehal:~$ docker version
Client:
 Version:           27.2.0
 API version:       1.47
 Go version:        go1.21.13
 Git commit:        3ab4256
 Built:             Tue Aug 27 14:14:20 2024
 OS/Arch:           linux/amd64
 Context:           default
```

```
myuser@snehal:~$ docker version
Client:
 Version:           27.2.0
 API version:       1.47
 Go version:        go1.21.13
 Git commit:        3ab4256
 Built:             Tue Aug 27 14:14:20 2024
 OS/Arch:           linux/amd64
 Context:           default

Server: Docker Desktop  ()
 Engine:
  Version:          27.2.0
  API version:      1.47 (minimum version 1.24)
  Go version:       go1.21.13
  Git commit:       3ab5c7d
  Built:            Tue Aug 27 14:15:15 2024
  OS/Arch:          linux/amd64
  Experimental:     false
 containerd:
  Version:          1.7.20
  GitCommit:        8fc6bcff51318944179630522a095cc9dbf9f353
 runc:
  Version:          1.1.13
  GitCommit:        v1.1.13-0-g58aa920
 docker-init:
  Version:          0.19.0
  GitCommit:        de40ad0
```

PHASE 2 — START THE FABRIC NETWORK

Open a **brand new terminal (Terminal 1)**. Run the two commands below and keep this terminal open for the entire experiment. Do NOT close it.

Step 8: Launch Microfab (Terminal 1)

code:
export MICROFAB_CONFIG='{"endorsing_organisations":[{"name":"Org1"}],"channels":[{"name":"mango-channel","endorsing_organisations":["Org1"]}],"capability_level":"V2_0"}'
code:
docker run -e MICROFAB_CONFIG -p 8080:8080 ibmcom/ibp-microfab
```
Wait until you see this line before continuing:
```
Microfab started in XXXX ms

```
myuser@snehal:~$ export MICROFAB_CONFIG='{
  "port": 8080,
  "endorsing_organizations": [
    {
      "name": "ProducersOrg"
    },
    {
      "name": "SellersOrg"
    }
  ],
  "channels": [
    {
      "name": "mango-channel",
      "endorsing_organizations": [
        "ProducersOrg",
        "SellersOrg"
      ]
    }
  ]
}'
```

```
myuser@snehal:~$ docker run -e MICROFAB_CONFIG -p 8080:8080 ibmcom/ibp-microfab
Unable to find image 'ibmcom/ibp-microfab:latest' locally
latest: Pulling from ibmcom/ibp-microfab
78f82eb67606: Download complete
b336a633f9e2: Download complete
b25deee7e50f: Download complete
9041dff50abb: Download complete
f24a19b34ee1: Download complete
c50c12d7db31: Download complete
576db8757925: Download complete
5186d5863148: Download complete
734dce7f474b: Download complete
eb98c0bd8a72: Download complete
65606c3065d6: Download complete
690939f6038f: Download complete
2785a9abab40: Download complete
d6e3788a121c: Download complete
531dcea8d07e: Download complete
Digest: sha256:125295f59f5be6c719141999a42b584ba5dcf7e388aa51c2c8d12cb2d2ca5859
Status: Downloaded newer image for ibmcom/ibp-microfab:latest
[       microfabd] 2026/04/03 13:30:49 Starting Microfab ...
[       microfabd] 2026/04/03 13:30:49 MBW
```

Hyperledger Fabric network is fully bootstrapped

```
[  sellersorgpeer] 2026-04-03 13:30:51.921 UTC 003a INFO [gossip.comm] GossipStream -> Peer 576d81a0d8d
48ce669212ad1153a4e1c115ae17edcedddad54007509f4c7dc9 ([::1]:59946) probed us
[  sellersorgpeer] 2026-04-03 13:30:51.921 UTC 003b INFO [comm.grpc.server] 1 -> streaming call complete
d grpc.service=gossip.Gossip grpc.method=GossipStream grpc.request_deadline=2026-04-03T13:31:01.918Z gr
c.peer_address=[::1]:59946 grpc.code=OK grpc.call_duration=2.792718ms
[  sellersorgpeer] 2026-04-03 13:30:51.939 UTC 003c INFO [comm.grpc.server] 1 -> unary call completed g
pc.service=gossip.Gossip grpc.method=Ping grpc.request_deadline=2026-04-03T13:30:53.939Z grpc.peer_addr
ss=[::1]:59946 grpc.code=OK grpc.call_duration=111.991µs
[producersorgpeer] 2026-04-03 13:30:52.024 UTC 0039 INFO [comm.grpc.server] 1 -> unary call completed g
pc.service=gossip.Gossip grpc.method=Ping grpc.request_deadline=2026-04-03T13:30:54.023Z grpc.peer_addr
ss=[::1]:49786 grpc.code=OK grpc.call_duration=131.73µs
[producersorgpeer] 2026-04-03 13:30:52.027 UTC 003a INFO [gossip.comm] GossipStream -> Peer e5edb156072
b17aa870ed0aff036986f7434f0a987d18c3655973758b754b12 ([::1]:49786) probed us
[producersorgpeer] 2026-04-03 13:30:52.027 UTC 003b INFO [comm.grpc.server] 1 -> streaming call complete
d grpc.service=gossip.Gossip grpc.method=GossipStream grpc.request_deadline=2026-04-03T13:31:02.026Z gr
c.peer_address=[::1]:49786 grpc.code=OK grpc.call_duration=1.324849ms
[  sellersorgpeer] 2026-04-03 13:30:55.832 UTC 003d INFO [gossip.channel] reportMembershipChanges -> [[
ango-channel] Membership view has changed. peers went online:  [[producersorgpeer-api.127-0-0-1.nip.io:
080 ]] , current view:  [[producersorgpeer-api.127-0-0-1.nip.io:8080 ]]]
[producersorgpeer] 2026-04-03 13:30:55.839 UTC 003c INFO [gossip.channel] reportMembershipChanges -> [[
ango-channel] Membership view has changed. peers went online:  [[sellersorgpeer-api.127-0-0-1.nip.io:80
0 ]] , current view:  [[sellersorgpeer-api.127-0-0-1.nip.io:8080 ]]]
```

PHASE 3 — PROJECT SETUP

Step 9: Create project directory

code:
cd ~
mkdir KBA-Mango
cd KBA-Mango

Step 10: Download Hyperledger Fabric binaries

code:
curl -sSL https://raw.githubusercontent.com/hyperledger/fabric/main/scripts/install-fabric.sh |
code: -s -- binary

Wait for it to finish. Then verify:

code:
ls bin/

Expected: you should see files like peer, orderer, fabric-ca-client etc.

```
root@snehal:~# su - myuser
myuser@snehal:~$ cd ~
mkdir KBA-Mango
cd KBA-Mango
myuser@snehal:~/KBA-Mango$
```

```
myuser@snehal:~/KBA-Mango$ curl -sSL https://raw.githubusercontent.com/hyper
ledger/fabric/main/scripts/install-fabric.sh | bash -s -- binary

Pull Hyperledger Fabric binaries

===> Downloading version 2.5.15 platform specific fabric binaries
===> Downloading:  https://github.com/hyperledger/fabric/releases/download/v
2.5.15/hyperledger-fabric-linux-amd64-2.5.15.tar.gz
===> Will unpack to: /home/myuser/KBA-Mango
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Curr
ent
                                 Dload  Upload   Total   Spent    Left  Spee
d
  0     0    0     0    0     0      0      0 --:--:-- --:--:-- --:--:--
  0     0    0     0    0     0      0      0 --:--:-- --:--:-- --:--:--
  0     0    0     0    0     0      0      0 --:--:-- --:--:-- --:--:--
 0
  0  120M    0 32750    0     0  20612      0  1:42:35  0:00:01  1:42:34 206
  7  120M    7 9487k    0     0  4113k      0  0:00:30  0:00:02  0:00:28 12.
 17  120M   17 21.4M    0     0  6637k      0  0:00:18  0:00:03  0:00:15 12.
 27  120M   27 33.3M    0     0  7910k      0  0:00:15  0:00:04  0:00:11 12.
 33  120M   33 40.0M    0     0  7700k      0  0:00:16  0:00:05  0:00:11 10.
 41  120M   41 50.0M    0     0  7751k      0  0:00:15  0:00:06  0:00:09  9.
 48  120M   48 58.9M    0     0  8262k      0  0:00:14  0:00:07  0:00:07  9.
 59  120M   59 71.4M    0     0  8806k      0  0:00:14  0:00:08  0:00:06  9.
 69  120M   69 84.0M    0     0  9243k      0  0:00:13  0:00:09  0:00:04 10.
 79  120M   79 96.6M    0     0  9599k      0  0:00:12  0:00:10  0:00:02 11.
 90  120M   90  109M    0     0  9890k      0  0:00:12  0:00:11  0:00:01 12.
 95  120M   95  114M    0     0  9565k      0  0:00:12  0:00:12 --:--:-- 11.
100  120M  100  120M    0     0  9675k      0  0:00:12  0:00:12 --:--:-- 11.
0M
==> Done.
===> Downloading version 1.5.17 platform specific fabric-ca-client binary
```

```
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Curr
ent
                                 Dload  Upload   Total   Spent    Left  Spee
d
  0     0    0     0    0     0      0      0 --:--:-- --:--:-- --:--:--
  0     0    0     0    0     0      0      0 --:--:-- --:--:-- --:--:--
 0
  0     0    0     0    0     0      0      0 --:--:-- --:--:-- --:--:--
  9 31.6M    9 3199k    0     0  2163k      0  0:00:14  0:00:01  0:00:13 323
 36 31.6M   36 11.4M    0     0  4714k      0  0:00:06  0:00:02  0:00:04 587
 74 31.6M   74 23.4M    0     0  6897k      0  0:00:04  0:00:03  0:00:01 802
100 31.6M  100 31.6M    0     0  7455k      0  0:00:04  0:00:04 --:--:-- 840
0k
==> Done.
```

```
myuser@snehal:~/KBA-Mango$ ls bin/
configtxgen    cryptogen  fabric-ca-client  ledgerutil  osnadmin
configtxlator  discover   fabric-ca-server  orderer     peer
```

Step 11: Set all environment variables

Run all of these one by one:

code:
export PATH=$PATH:${PWD}/bin
export FABRIC_CFG_PATH=${PWD}/config
export GOPATH=$HOME/go
export PATH=$PATH:$GOPATH/bin

Verify peer binary works:

code:
peer version

```
myuser@snehal:~/KBA-Mango$ export PATH=$PATH:${PWD}/bin
export FABRIC_CFG_PATH=${PWD}/config
export GOPATH=$HOME/go
export PATH=$PATH:$GOPATH/bin
```

```
myuser@snehal:~/KBA-Mango$ peer version
peer:
 Version: v2.5.15
 Commit SHA: 83c7930
 Go version: go1.26.0
 OS/Arch: linux/amd64
 Chaincode:
  Base Docker Label: org.hyperledger.fabric
  Docker Namespace: hyperledger
```

Step 12: Generate wallets, gateway, and MSP

code:
curl -s http://console.127-0-0-1.nip.io:8080/ak/api/v1/components | weft microfab -w ./_wallets -p ./_gateways -m ./_msp -f
```
You will see output like this — copy the `export` lines shown for `org1admin`:
```
For org1admin @ Org1 use these:

export CORE_PEER_LOCALMSPID=Org1MSP
export CORE_PEER_MSPCONFIGPATH=/home/<user>/KBA-Mango/_msp/Org1/org1admin/msp
export CORE_PEER_ADDRESS=org1peer-api.127-0-0-1.nip.io:8080

```
myuser@snehal:~/KBA-Mango$ curl -s http://console.127-0-0-1.nip.io:8080/ak/a
pi/v1/components | weft microfab -w ./_wallets -p ./_gateways -m ./_msp -f
Gateway profile written to : /home/myuser/KBA-Mango/_gateways/producersorgga
teway.json
Gateway profile written to : /home/myuser/KBA-Mango/_gateways/sellersorggate
way.json
Added identity under label sellersorgadmin to the wallet at /home/myuser/KBA
-Mango/_wallets/SellersOrg
Added identity under label sellersorgcaadmin to the wallet at /home/myuser/K
BA-Mango/_wallets/SellersOrg
Added identity under label ordereradmin to the wallet at /home/myuser/KBA-Ma
ngo/_wallets/Orderer
```

```
For producersorgcaadmin @  ProducersOrg use these:

export CORE_PEER_LOCALMSPID=ProducersOrgMSP
export CORE_PEER_MSPCONFIGPATH=/home/myuser/KBA-Mango/_msp/ProducersOrg/prod
ucersorgcaadmin/msp
export CORE_PEER_ADDRESS=producersorgpeer-api.127-0-0-1.nip.io:8080
```

```
For sellersorgadmin @  SellersOrg use these:

export CORE_PEER_LOCALMSPID=SellersOrgMSP
export CORE_PEER_MSPCONFIGPATH=/home/myuser/KBA-Mango/_msp/SellersOrg/seller
sorgadmin/msp
export CORE_PEER_ADDRESS=sellersorgpeer-api.127-0-0-1.nip.io:8080

For sellersorgcaadmin @  SellersOrg use these:

export CORE_PEER_LOCALMSPID=SellersOrgMSP
export CORE_PEER_MSPCONFIGPATH=/home/myuser/KBA-Mango/_msp/SellersOrg/seller
sorgcaadmin/msp
export CORE_PEER_ADDRESS=sellersorgpeer-api.127-0-0-1.nip.io:8080
```

Step 13: Set peer environment variables

Run exactly the three lines shown in the output above. They will look like this (use YOUR actual username, not the example):

code:
export CORE_PEER_LOCALMSPID=Org1MSP
export
CORE_PEER_MSPCONFIGPATH=/home/<your-username>/KBA-Mango/_msp/Org1/org1admi
n/msp
export CORE_PEER_ADDRESS=org1peer-api.127-0-0-1.nip.io:8080

If you don't know your username:

```
myuser@snehal:~/KBA-Mango$ export CORE_PEER_LOCALMSPID=ProducersOrgMSP
export CORE_PEER_MSPCONFIGPATH=/home/myuser/KBA-Mango/_msp/ProducersOrg/produ
cersorgadmin/msp
export CORE_PEER_ADDRESS=producersorgpeer-api.127-0-0-1.nip.io:8080
myuser@snehal:~/KBA-Mango$
```

code:
whoami

Verify the MSP folder exists:

code:
ls ./_msp/Org1/org1admin/msp

Expected: admincerts cacerts keystore signcerts

```
myuser@snehal:~/KBA-Mango$ ls ./_msp/ProducersOrg/producersorgadmin/msp
admincerts  cacerts  keystore  signcerts
```

PHASE 4 — WRITE THE CHAINCODE

Step 14: Create the chaincode directory and initialise Go module

code:
cd ~/KBA-Mango
mkdir KBA-Mango-Contract
cd KBA-Mango-Contract
go mod init mango

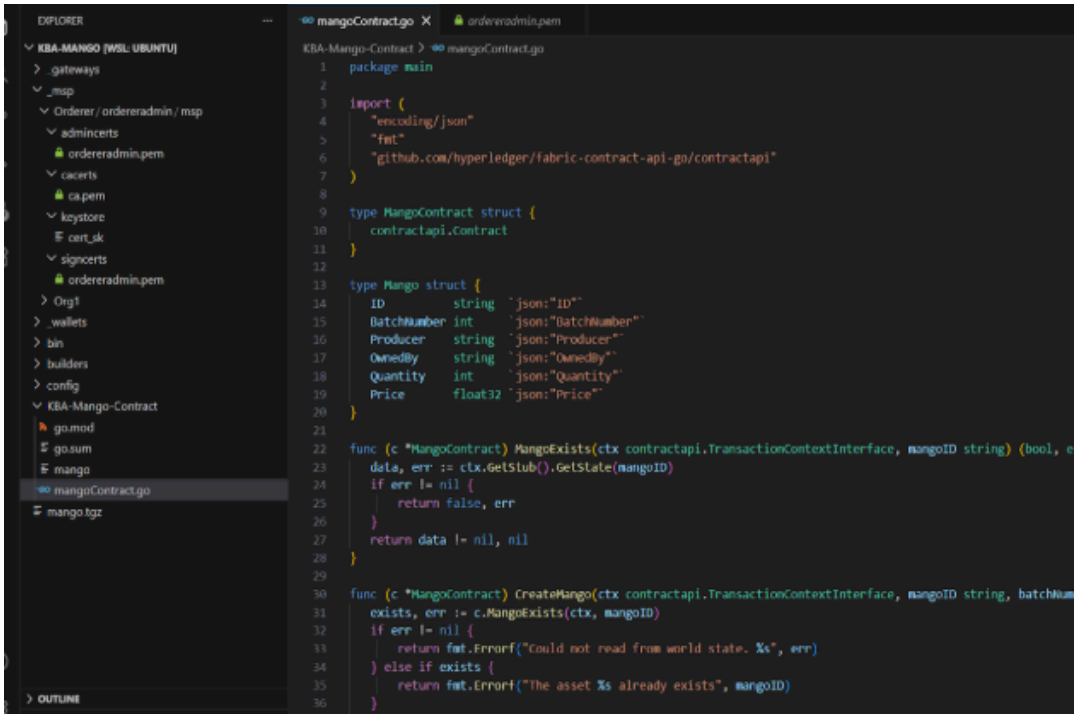
Create the Chaincode File

touch mangoContract.go
nano mangoContract.go

```
myuser@snehal:~/KBA-Mango$ cd ~/KBA-Mango
mkdir KBA-Mango-Contract
cd KBA-Mango-Contract
```

```
myuser@snehal:~/KBA-Mango/KBA-Mango-Contract$ go mod init mango
go: creating new go.mod: module mango
```

```
myuser@snehal:~/KBA-Mango/KBA-Mango-Contract$ go get github.com/hyperledger/
fabric-contract-api-go@v1.2.1
go: downloading github.com/hyperledger/fabric-contract-api-go v1.2.1
go: added github.com/hyperledger/fabric-contract-api-go v1.2.1
myuser@snehal:~/KBA-Mango/KBA-Mango-Contract$ go get google.golang.org/grpc@
v1.56.3
go: downloading google.golang.org/grpc v1.56.3
go: downloading golang.org/x/net v0.9.0
go: downloading github.com/golang/protobuf v1.5.3
```



Step 15: Write the chaincode file

Paste the complete chaincode below using Ctrl+Shift+V. Then save with Ctrl+O, Enter. Exit with Ctrl+X.

code:

```
package main

import (
	"encoding/json"
	"fmt"
	"github.com/hyperledger/fabric-contract-api-go/contractapi"
)

// MangoContract defines the smart contract structure
type MangoContract struct {
	contractapi.Contract
}

// Mango defines the asset structure stored on the ledger
type Mango struct {
	ID          string  `json:"ID"`
	BatchNumber int     `json:"BatchNumber"`
	Producer    string  `json:"Producer"`
```

```
	OwnedBy     string  `json:"OwnedBy"`
	Quantity    int     `json:"Quantity"`
	Price       float32 `json:"Price"`
}

// MangoExists checks if a mango asset exists in the world state
func (c *MangoContract) MangoExists(ctx contractapi.TransactionContextInterface, mangoID
string) (bool, error) {
	data, err := ctx.GetStub().GetState(mangoID)
	if err != nil {
		return false, err
	}
	return data != nil, nil
}

// CreateMango adds a new mango asset to the ledger
func (c *MangoContract) CreateMango(ctx contractapi.TransactionContextInterface, mangoID
string, batchNumber int, producer string, quantity int, price float32) error {
	exists, err := c.MangoExists(ctx, mangoID)
	if err != nil {
		return fmt.Errorf("Could not read from world state. %s", err)
	} else if exists {
		return fmt.Errorf("The asset %s already exists", mangoID)
	}
	mango := Mango{
		ID:          mangoID,
		BatchNumber: batchNumber,
		Producer:    producer,
		OwnedBy:     producer,
		Quantity:    quantity,
		Price:       price,
	}
	bytes, _ := json.Marshal(mango)
	return ctx.GetStub().PutState(mangoID, bytes)
}

// ReadMango retrieves a mango asset from the ledger
func (c *MangoContract) ReadMango(ctx contractapi.TransactionContextInterface, mangoID
string) (*Mango, error) {
	exists, err := c.MangoExists(ctx, mangoID)
	if err != nil {
		return nil, fmt.Errorf("Could not read from world state. %s", err)
	} else if !exists {
		return nil, fmt.Errorf("The asset %s does not exist", mangoID)
```

```
	}
	bytes, _ := ctx.GetStub().GetState(mangoID)
	mango := new(Mango)
	err = json.Unmarshal(bytes, mango)
	if err != nil {
		return nil, fmt.Errorf("Could not unmarshal world state data to type Mango")
	}
	return mango, nil
}

// UpdateMango updates an existing mango asset on the ledger
func (c *MangoContract) UpdateMango(ctx contractapi.TransactionContextInterface, mangoID
string, batchNumber int, producer string, owner string, quantity int, price float32) error {
	exists, err := c.MangoExists(ctx, mangoID)
	if err != nil {
		return fmt.Errorf("Could not read from world state. %s", err)
	} else if !exists {
		return fmt.Errorf("The asset %s does not exist", mangoID)
	}
	mango := Mango{
		ID:          mangoID,
		BatchNumber: batchNumber,
		Producer:    producer,
		OwnedBy:     owner,
		Quantity:    quantity,
		Price:       price,
	}
	bytes, _ := json.Marshal(mango)
	return ctx.GetStub().PutState(mangoID, bytes)
}

// DeleteMango removes a mango asset from the ledger
func (c *MangoContract) DeleteMango(ctx contractapi.TransactionContextInterface, mangoID
string) error {
	exists, err := c.MangoExists(ctx, mangoID)
	if err != nil {
		return fmt.Errorf("Could not read from world state. %s", err)
	} else if !exists {
		return fmt.Errorf("The asset %s does not exist", mangoID)
	}
	return ctx.GetStub().DelState(mangoID)
}

// SellMango transfers ownership of a mango asset to a new owner
```

```
func (c *MangoContract) SellMango(ctx contractapi.TransactionContextInterface, mangoID
string, owner string, newOwner string) error {
	exists, err := c.MangoExists(ctx, mangoID)
	if err != nil {
		return fmt.Errorf("Could not read from world state. %s", err)
	} else if !exists {
		return fmt.Errorf("The asset %s does not exist", mangoID)
	}
	mango, _ := c.ReadMango(ctx, mangoID)
	if mango.OwnedBy == owner {
		mango.OwnedBy = newOwner
		bytes, _ := json.Marshal(mango)
		return ctx.GetStub().PutState(mangoID, bytes)
	} else {
		return fmt.Errorf("Asset is not owned by %v, only original owner can sell the
asset", owner)
	}
}

func main() {
	mangoContract := new(MangoContract)
	chaincode, err := contractapi.NewChaincode(mangoContract)
	if err != nil {
		panic("Could not create chaincode." + err.Error())
	}
	err = chaincode.Start()
	if err != nil {
		panic("Failed to start chaincode. " + err.Error())
	}
}
```

PHASE 5 — FIX DEPENDENCIES

Microfab builds chaincode internally using Go 1.18. Default pulled dependencies require Go 1.19+. These exact pinned versions are the ones that work.

Step 16: Pull the main fabric dependency first

code:
go get github.com/hyperledger/fabric-contract-api-go@v1.2.1

Step 17: Pin gRPC to a Go 1.18 compatible version

code:
go get google.golang.org/grpc@v1.56.3

```
myuser@snehal:~/KBA-Mango/KBA-Mango-Contract$ go get github.com/go-openapi/
sonpointer@v0.19.6
go get github.com/go-openapi/jsonreference@v0.20.2
go get github.com/go-openapi/swag@v0.22.4
go: downloading github.com/go-openapi/jsonpointer v0.19.6
go: downloading github.com/go-openapi/swag v0.22.3
go: downloading github.com/mailru/easyjson v0.7.7
go: downloading gopkg.in/yaml.v3 v3.0.1
go: downloading github.com/josharian/intern v1.0.0
go: upgraded github.com/go-openapi/jsonpointer v0.19.5 => v0.19.6
go: upgraded github.com/go-openapi/swag v0.21.1 => v0.22.3
go: downloading github.com/go-openapi/jsonreference v0.20.2
```

Step 18: Pin openapi dependencies to Go 1.18 compatible versions

code:
go get github.com/go-openapi/jsonpointer@v0.19.6
go get github.com/go-openapi/jsonreference@v0.20.2
go get github.com/go-openapi/swag@v0.22.4


```
myuser@snehal:~/KBA-Mango/KBA-Mango-Contract$ go get github.com/go-openapi/
sonpointer@v0.19.6
go get github.com/go-openapi/jsonreference@v0.20.2
go get github.com/go-openapi/swag@v0.22.4
go: downloading github.com/go-openapi/jsonpointer v0.19.6
go: downloading github.com/go-openapi/swag v0.22.3
go: downloading github.com/mailru/easyjson v0.7.7
go: downloading gopkg.in/yaml.v3 v3.0.1
go: downloading github.com/josharian/intern v1.0.0
go: upgraded github.com/go-openapi/jsonpointer v0.19.5 => v0.19.6
go: upgraded github.com/go-openapi/swag v0.21.1 => v0.22.3
go: downloading github.com/go-openapi/jsonreference v0.20.2
```

Step 19: Tidy and build

code:
go mod tidy
go build ./…

```
myuser@snehal:~/KBA-Mango/KBA-Mango-Contract$ go mod tidy
go build ./...
go: downloading github.com/hyperledger/fabric-protos-go v0.3.0
go: downloading github.com/hyperledger/fabric-chaincode-go v0.0.0-2023022819
4215-b84622ba6a7a
go: downloading github.com/stretchr/testify v1.8.2
go: downloading github.com/go-openapi/spec v0.20.8
```

```
myuser@snehal:~/KBA-Mango/KBA-Mango-Contract$ go mod tidy
go build ./...
go: downloading github.com/hyperledger/fabric-protos-go v0.3.0
go: downloading github.com/hyperledger/fabric-chaincode-go v0.0.0-2023022819
4215-b84622ba6a7a
go: downloading github.com/stretchr/testify v1.8.2
go: downloading github.com/go-openapi/spec v0.20.8
go: downloading github.com/xeipuuv/gojsonschema v1.2.0
go: downloading github.com/gobuffalo/packr v1.30.1
go: downloading gopkg.in/yaml.v2 v2.4.0
go: downloading gopkg.in/check.v1 v1.0.0-20201130134442-10cb98267c6c
```

PHASE 6 — PACKAGE AND INSTALL CHAINCODE

Step 20: Go back to the project root

code:
cd ~/KBA-Mango

Step 21: Package the chaincode

code:
peer lifecycle chaincode package mango.tgz \
  --path ./KBA-Mango-Contract/ \
  --lang golang \
  --label mango_1

Verify the package was created:

```
myuser@snehal:~/KBA-Mango/KBA-Mango-Contract$ cd ~/KBA-Mango
myuser@snehal:~/KBA-Mango$ peer lifecycle chaincode package mango.tgz \
--path ./KBA-Mango-Contract/ \
--lang golang \
--label mango_1
```

Step 22: Install chaincode on the peer

code:
peer lifecycle chaincode install mango.tgz

Expected output:

INFO Installed remotely: response:<status:200 ...>
INFO Chaincode code package identifier: mango_1:<long-hash-string>

> Copy the entire mango_1:xxxxxx... identifier from the output. You need it in the next step.

```
myuser@snehal:~/KBA-Mango$ peer lifecycle chaincode install mango.tgz
2026-04-03 16:53:43.757 UTC 0001 INFO [cli.lifecycle.chaincode] submitInstal
lProposal -> Installed remotely: response:<status:200 payload:"\nHmango_1:01
dc00b9bf1607543e76c991bff583d23847a19fc066071c0cff6871494cda2b\022\007mango_
1" >
2026-04-03 16:53:43.758 UTC 0002 INFO [cli.lifecycle.chaincode] submitInstal
lProposal -> Chaincode code package identifier: mango_1:01dc00b9bf1607543e76
c991bff583d23847a19fc066071c0cff6871494cda2b
```

```
myuser@snehal:~/KBA-Mango$ peer lifecycle chaincode approveformyorg \
-o orderer-api.127-0-0-1.nip.io:8080 \
--channelID mango-channel \
--name mango \
--version 1 \
--sequence 1 \
--waitForEvent \
--package-id ${CC_PACKAGE_ID}
2026-04-03 16:59:32.943 UTC 0001 INFO [chaincodeCmd] ClientWait -> txid [319
ae6a457d5dbce7a07a14e63ae730f30454882597352a3e172dc55601f4ab2] committed wit
h status (VALID) at producersorgpeer-api.127-0-0-1.nip.io:8080
```

Step 23: Export the Package ID

Paste your exact identifier from the output above:

code:
export CC_PACKAGE_ID=mango_1:<paste-your-exact-hash-here>

Verify it is set:

code:
echo $CC_PACKAGE_ID

Expected: prints your full package ID, not empty.

```
myuser@snehal:~/KBA-Mango$ export CC_PACKAGE_ID=mango_1:01dc00b9bf1607543e76
c991bff583d23847a19fc066071c0cff6871494cda2b
myuser@snehal:~/KBA-Mango$ peer lifecycle chaincode approveformyorg \
-o orderer-api.127-0-0-1.nip.io:8080 \
--channelID mango-channel \
--name mango \
--version 1 \
--sequence 1 \
--waitForEvent \
--package-id ${CC_PACKAGE_ID}
```

PHASE 7 — APPROVE AND COMMIT

Step 24: Approve chaincode for Org1

code:
```
peer lifecycle chaincode approveformyorg \
  -o orderer-api.127-0-0-1.nip.io:8080 \
  --channelID mango-channel \
  --name mango \
  --version 1 \
  --sequence 1 \
  --waitForEvent \
  --package-id ${CC_PACKAGE_ID}
```

Expected: txid [...] committed with status (VALID)


```
myuser@snehal:~/KBA-Mango$ export CC_PACKAGE_ID=mango_1:01dc00b9bf1607543e76
c991bff583d23847a19fc066071c0cff6871494cda2b
myuser@snehal:~/KBA-Mango$ peer lifecycle chaincode approveformyorg \
-o orderer-api.127-0-0-1.nip.io:8080 \
--channelID mango-channel \
--name mango \
--version 1 \
--sequence 1 \
--waitForEvent \
--package-id ${CC_PACKAGE_ID}
```

Step 25: Commit chaincode to the channel

code:
```
peer lifecycle chaincode commit \
  -o orderer-api.127-0-0-1.nip.io:8080 \
  --channelID mango-channel \
  --name mango \
  --version 1 \
  --sequence 1
```

Expected: txid [...] committed with status (VALID)

```
2026-04-03 16:59:32.943 UTC 0001 INFO [chaincodeCmd] ClientWait -> txid [319
ae6a457d5dbce7a07a14e63ae730f30454882597352a3e172dc55601f4ab2] committed wit
h status (VALID) at producersorgpeer-api.127-0-0-1.nip.io:8080
```

```
myuser@snehal:~/KBA-Mango$ peer lifecycle chaincode commit \
-o orderer-api.127-0-0-1.nip.io:8080 \
--channelID mango-channel \
--name mango \
--version 1 \
--sequence 1
2026-04-03 17:01:16.359 UTC 0001 INFO [chaincodeCmd] ClientWait -> txid [f9c
e54801ecb20d90d6e985e10d7dca2032c591c0b15b5f2907d3d4351a21a6a] committed wit
h status (VALID) at producersorgpeer-api.127-0-0-1.nip.io:8080
```

PHASE 8 — TEST ALL CHAINCODE FUNCTIONS

Step 26: CreateMango

code:
```
peer chaincode invoke \
  -o orderer-api.127-0-0-1.nip.io:8080 \
  --channelID mango-channel \
  -n mango \
  -c '{"function":"CreateMango","Args":["MANGO1","1","mango corp","5000","10000"]}'
```

Expected: Chaincode invoke successful. result: status:200

```
myuser@snehal:~/KBA-Mango$ peer chaincode invoke \
-o orderer-api.127-0-0-1.nip.io:8080 \
--channelID mango-channel \
-n mango \
-c '{"function":"CreateMango","Args":["MANGO1","1","mango corp","5000","1000
0"]}'
2026-04-03 17:03:26.033 UTC 0001 INFO [chaincodeCmd] chaincodeInvokeOrQuery
-> Chaincode invoke successful. result: status:200
```

Step 27: ReadMango

code:
peer chaincode query \
  -o orderer-api.127-0-0-1.nip.io:8080 \
  --channelID mango-channel \
  -n mango \
  -c '{"function":"ReadMango","Args":["MANGO1"]}'

Expected:

json
{"ID":"MANGO1","BatchNumber":1,"Producer":"mango corp","OwnedBy":"mango corp","Quantity":5000,"Price":10000}

```
myuser@snehal:~/KBA-Mango$ peer chaincode query \
-o orderer-api.127-0-0-1.nip.io:8080 \
--channelID mango-channel \
-n mango \
-c '{"function":"ReadMango","Args":["MANGO1"]}'
{"ID":"MANGO1","BatchNumber":1,"Producer":"mango corp","OwnedBy":"mango corp
","Quantity":5000,"Price":10000}
myuser@snehal:~/KBA-Mango$
```

Step 28: UpdateMango

code:
peer chaincode invoke \
  -o orderer-api.127-0-0-1.nip.io:8080 \
  --channelID mango-channel \
  -n mango \
  -c '{"function":"UpdateMango","Args":["MANGO1","2","mango corp","mango corp","6000","12000"]}'

Expected: Chaincode invoke successful. result: status:200

```
myuser@snehal:~/KBA-Mango$ peer chaincode invoke \
-o orderer-api.127-0-0-1.nip.io:8080 \
--channelID mango-channel \
-n mango \
-c '{"function":"UpdateMango","Args":["MANGO1","2","mango corp","mango corp"
,"6000","12000"]}'
2026-04-03 17:04:46.382 UTC 0001 INFO [chaincodeCmd] chaincodeInvokeOrQuery
-> Chaincode invoke successful. result: status:200
myuser@snehal:~/KBA-Mango$
```

Step 29: SellMango

code:
peer chaincode invoke \
  -o orderer-api.127-0-0-1.nip.io:8080 \
  --channelID mango-channel \
  -n mango \
  -c '{"function":"SellMango","Args":["MANGO1","mango corp","mango shop"]}'

Expected: Chaincode invoke successful. result: status:200

```
myuser@snehal:~/KBA-Mango$ peer chaincode invoke \
-o orderer-api.127-0-0-1.nip.io:8080 \
--channelID mango-channel \
-n mango \
-c '{"function":"SellMango","Args":["MANGO1","mango corp","mango shop"]}'
2026-04-03 17:05:16.620 UTC 0001 INFO [chaincodeCmd] chaincodeInvokeOrQuery
-> Chaincode invoke successful. result: status:200
myuser@snehal:~/KBA-Mango$
```

Step 30: ReadMango — verify ownership transferred

code:
peer chaincode query \
  -o orderer-api.127-0-0-1.nip.io:8080 \
  --channelID mango-channel \
  -n mango \
  -c '{"function":"ReadMango","Args":["MANGO1"]}'

Expected:

{"ID":"MANGO1","BatchNumber":2,"Producer":"mango corp","OwnedBy":"mango shop","Quantity":6000,"Price":12000}

```
myuser@snehal:~/KBA-Mango$ peer chaincode query \
-c '{"function":"ReadMango","Args":["MANGO1"]}'
Error: The required parameter 'channelID' is empty. Rerun the command with -
C flag
Usage:
  peer chaincode query [flags]

Flags:
  -C, --channelID string               The channel on which this command sho
uld be executed
      --connectionProfile string       Connection profile that provides the
necessary connection information for the network. Note: currently only suppo
rted for providing peer connection information
  -c, --ctor string                    Constructor message for the chaincode
 in JSON format (default "{}")
  -h, --help                           help for query
  -x, --hex                            If true, output the query value byte
array in hexadecimal. Incompatible with --raw
  -n, --name string                    Name of the chaincode
      --peerAddresses stringArray      The addresses of the peers to connect
 to
```

Step 31: DeleteMango

code:
peer chaincode invoke \
  -o orderer-api.127-0-0-1.nip.io:8080 \
  --channelID mango-channel \
  -n mango \
  -c '{"function":"DeleteMango","Args":["MANGO1"]}'
```
Expected: `Chaincode invoke successful. result: status:200`

```
myuser@snehal:~/KBA-Mango$ peer chaincode invoke \
-o orderer-api.127-0-0-1.nip.io:8080 \
--channelID mango-channel \
-n mango \
-c '{"function":"DeleteMango","Args":["MANGO1"]}'
2026-04-03 17:06:09.975 UTC 0001 INFO [chaincodeCmd] chaincodeInvokeOrQuery
-> Chaincode invoke successful. result: status:200
```
```

PHASE 9 — CLEANUP

Step 32: Stop Microfab

Go to Terminal 1 and press `Ctrl+C`. Wait for:

Microfab stopped

Step 33: Remove stopped containers

code:
docker container prune

Type y and press Enter.

```
myuser@snehal:~$ docker container prune
WARNING! This will remove all stopped containers.
Are you sure you want to continue? [y/N] y
Deleted Containers:
df15175e23e884122621fe73740e055cc6d911262859cb7d704780b504303fc9
5a38a7d6fceb559c8c5fef1c17b92c9e32879289dddbf7643966459f0c61ff5c
```